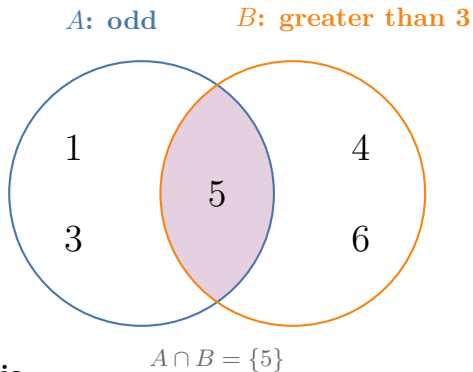


Adding $P(A)$ and $P(B)$ counts the overlap twice



$$P(A) = 3/6, \quad P(B) = 3/6$$

$3/6 + 3/6 = 6/6$: too much, 5 counted twice

$$P(A \cup B) = 3/6 + 3/6 - 1/6 = 5/6$$